

Executive summary — Visual Augmentation of Audio Semantics for Accessibility

*For someone stepping into this project today (updated 2026-09-15, evening).
Written in plain language; every technical word is explained the first time it appears. Every number here comes from a file in the repository. The full record is in `docs/project_notes.tex` (the dated lab notebook), `docs/report/report.pdf` (the thesis draft, 15 pages) and `LIMITATIONS.md` (what the numbers do not support).*

1. The problem

A deaf viewer gets a video's speech through subtitles, but not the rest of the soundtrack: a siren behind the camera, a dog barking in the next room, glass breaking off screen. Subtitle guidelines even tell captioners to *skip* sounds the picture already shows, and in practice most ambient sound is never captioned at all.

This project built a system that listens to a video, and for each ambient sound decides whether the thing making it is **already visible on screen**. If it is not, the system shows a small picture of that sound next to the video for as long as the sound lasts. If it is (you can see the dog barking), it shows nothing. That "show only when not visible" rule is called the **gate**. It is the whole idea, and the thing the evaluation tests.

Rules set by the supervisor: **no training** — the system is built by connecting existing, freely available AI models; **minimum human involvement** — decisions are made by models, not by hand-written rules; finish within the week.

2. Background: what already exists

- **Subtitles** carry speech well and ambient sound badly. **Sign language** is richer but rare and cannot be automated.
- **Sound-awareness apps for deaf users** (HomeSound, SoundWatch) recognise ~20 everyday sounds in the home — a knock, an alarm, a crying baby — and users prefer pictures to text. They are for real life, not for watching video.
- **Sound detectors**: AI models trained on *AudioSet* (a Google collection of 2

million YouTube clips labelled with 527 kinds of sound) can name what is in a soundtrack. We use one called **BEATs**.

- **Vision-language models (VLMs)**: AI models you can show pictures to and ask questions in words ("what is making this sound?"). We use **Qwen2.5-VL-7B** ("7B" = seven billion parameters, a mid-size model).
- **Audio-to-image systems** (e.g. Sound2Scene) turn a sound directly into a picture, without looking at the video. They are our main comparison: "draw everything you hear".
- We found no existing system that does all three: detect ambient sound → decide whether it is heard-but-not-seen → show a picture. That combination, plus a way to test it, is the contribution.

3. How the system works (seven stages, code in `src/`)

stage	what it does	model
1	pulls the audio out of the video	FFmpeg
2	quick list of objects seen anywhere in the clip (stage 5 makes the real decision)	OWLv2
3	writes down any speech — used only as a hint, never drawn	Whisper
4	detects ambient sounds and when they start and stop	BEATs
5	reasons : is the source visible? what should the picture show? which sounds are one source?	Qwen2.5-VL-7B
6	draws the picture and places it beside the video	FLUX.1 (an image generator)
7	evaluates the result	see §6

The important behaviours (all decided by models at run time; settings in `config.py`)

Detection (stage 4). BEATs scores every 2-second slice of audio, sliding 0.25 s at a time, giving each of the 527 sound kinds a confidence between 0 and 1. A sound counts if its confidence reaches **0.35** (the "bar"). Its start is stamped where the confidence begins to rise, not where it crosses the bar, so the picture is not late (this cut timing error from 0.43 s to 0.31 s on an outside dataset). Speech and music are never shown.

The visibility decision (stage 5, the gate). For each detected sound:

1. Its duration is cut into pieces of at most 5 seconds. For each piece, 6 video frames are taken, from 1 s before to 1 s after it.
2. The VLM is asked three differently worded questions about those frames: - *Name it*: "What in these frames is making the sound of X, or nothing?" If it names something, a follow-up checks common sense: "Does a make an X sound?" — so a cowboy hat cannot be the source of a vehicle sound. - *Is it happening*: "Can you SEE X happening — the source in frame and visibly making that sound?" Two options, asked twice with the options swapped; counted only if both answers agree (the model otherwise tends to pick the second option whatever it says). - *Describe*: "Describe what is in these frames." If the description names the source, or common sense says what it describes makes that sound, that counts as a vote.
3. Majority of the three votes = "visible" for that piece. The sound is silenced only if it is visible in **every** piece; otherwise the picture stays for the whole sound (one sound, one continuous picture — no flicker).

"Visible" means *seen making the sound*: a baby held in its mother's arms is not visibly crying; a fire-alarm box on a wall is not visibly ringing; a woman with her head back is visibly laughing.

What the picture shows. The VLM writes a 2–5-word phrase describing the *event* — "glass shattering", "audience clapping" — from the detector's label, the frames (a window or a bottle?) and a short place name ("kitchen"). Dialogue is never used (it once leaked into pictures). A second question checks the phrase points back to the right sound. FLUX draws it on a white background; the panel holds at most three pictures at once, each for at least 1.5 s.

One picture per sound source. The detector often gives one sound several names (Laughter, Giggle, Snicker). AudioSet's own family tree (Giggle is a kind of Laughter) merges them; a text-similarity check catches paraphrases the tree cannot; and when two names have exactly the same start and end (a crying baby heard as "Sheep" and "Baby cry"), the VLM looks at the frames and picks one.

Speech as a hint. If people on the soundtrack react to a sound ("whoa, what's that?"), that sound gets priority when the panel is full. It never overrides the visibility decision.

4. Lessons from building it

Every early fix was made on the clip in front of us and broke on the next one. The rules that came out of that:

1. **Lists fail.** A hand-written list of 30 "visible objects" could never mark laughter as visible. Every list became a question to a model.
2. **Ask for a name, not a yes.** A yes/no question to a VLM gets agreement, not evidence. A name can be checked.
3. **Read what a model already gives you.** Deduplication was solved by AudioSet's family tree after three cleverer ideas failed.
4. **Set thresholds once, on separate clips.** Since 2026-09-13: settings are chosen on a *development split* (clips not used for the final score), every failure is logged (`docs/failure_catalogue.md`), and a change needs three cases of one pattern.
5. **Evaluation code produces plausible wrong numbers.** An empty result scored 2/4; a similarity returned 0.0 for everything after a library update; an audio model was never given its audio and answered "none" 100 times. Each was found by reading logs.
6. **Declare the pass bar before running.** Every experiment this week wrote its pass/fail rule into the code before the job was submitted (see §8).

5. The benchmark (`benchmark/tags.json`)

274 short clips, labelled by Adam into four scenarios: **unseen** (an ambient sound whose source is off screen — a picture is due), **mixed** (some sources visible, some not — a picture is due), **seen** (the source is on screen — no picture due), **no ambient** (speech or music only — no picture due). The labelling rule is the one professional captioners use: "does the picture already tell the viewer this sound is happening?"

Test set: 100 clips, 25 per scenario — every reported score uses these.

Development split: the other 174 clips, used only for choosing settings. (All 25 mixed clips ended up in the test set; this mattered once, §8.)

Label reliability: Adam re-labelled 60 clips blind and agreed with himself on 78% ($\kappa = 0.60$, "moderate"); the disagreement is mostly on the "unseen" clips, which are exactly the ones the gate is judged on. One annotator; a second would make this firmer.

Outside check — DCASE 2025. A public research dataset of 30,000 five-second indoor clips in which humans marked, for every sound, when it starts and whether

its source is inside the camera's view. We use it as ground truth we did not make ourselves: on 258 of its sounds our visibility decision agrees 66.7% — it keeps 97% of off-screen sounds but silences only 35% of on-screen ones (it errs on the safe side: better a redundant picture than a missing one).

6. How the system is scored

There is no user study; scoring is automatic (as the proposal specified). For each clip and each system:

1. The system produces its output — pictures beside the video, or nothing.
2. A VLM looks at the result and writes what a viewer would learn from the pictures.
3. A **reference sentence** states what a hearing viewer gets from the soundtrack that a deaf viewer would miss — or "nothing beyond the picture" if nothing is missing.
4. A **judge** (a text model, Mistral-7B) scores how well (2) matches (3), from 0 to 4. Two cases are decided by code: showing nothing when something was missing = **0**; showing nothing when nothing was missing = **4**.

Two baselines are scored the same way: **blind** — draw a picture for every sound heard, never look at the video; **caption** — write the sound's name as text instead of a picture.

Who writes the reference sentence matters. Three versions:

- *model-derived* (the proposal's): written by the system's own models from its own detections. Found to be circular — it rewards whichever system repeats the detector.
- *human-grounded*: the same, corrected by Adam's labels (on seen / no-ambient clips it becomes "nothing beyond the picture"). **This is the one the headline uses.**
- *independent*: written by a different set of models that never saw our system. It fixes the circularity, but its own guess at "is the source visible?" is no better than a coin flip (55% agreement with the labels), so it cannot score the gate. Reported, not used.

The score has a built-in asymmetry. A picture wrongly withheld scores 0 where it could have scored 4 — a loss of ~4. A redundant picture (source already visible) scores ~3 where silence scores 4 — a loss of ~1. So the gate must be right about 80% of the time when it stays silent just to break even. This shapes every result

below.

Two things the judge cannot see (found 2026-09-15): if the detector hears *nothing* on a clip where a picture was due, the reference itself says nothing is missing and *every* system gets 4 — so a detector miss is invisible to the score. And the score cannot say whether a picture was shown at the right moment; only that it was shown.

7. Results (v3 = the current system; 100 test clips; score 0–4)

reference	gated (ours)	blind	caption	ours – blind
model-derived	2.64	3.04	2.79	–0.40 (clearly behind)
human-grounded	3.17	3.16	2.89	+0.01 — a tie (21 clips won, 64 tied, 15 lost)

Split by whether a picture was due (human-grounded):

- **no picture due** (50 clips): ours 3.56 vs blind 3.30 → **+0.26, we win** — we correctly stay quiet, blind draws a redundant picture.
- **picture due** (50 clips): ours 2.78 vs blind 3.02 → **–0.24, we lose** — sometimes we stay quiet when we should not.

In words: *the system does the right thing when nothing is missing, and still sometimes stays silent when something is*. A perfect gate would score 3.25. We beat the caption baseline (+0.28). The earlier version (v2, one week older) lost clearly under every reference; the week's changes closed the gap to a tie.

A second view of the same result. Because the score's 4-to-1 asymmetry comes from the rubric and not from deaf viewers, we also computed a simpler, judge-free score: per clip, did the gate open when a picture was due and stay shut when not? Ours: 39 hits, 11 misses, 28 redundant. Blind: 43 hits, 7 misses, 39 redundant. If a redundant picture is treated as costing a quarter of a miss (the judge's implicit rate), the two tie; if it costs 40% or more, ours is ahead. So: *the gate is preferred as soon as a redundant picture is counted as more than about 40% as bad as a missing one*. Whether real viewers feel it that way is an open question — no user study.

Where the remaining errors come from (read from saved decisions, no new runs):

- Of the 11 test clips where we stayed silent and should not have: **7 are the**

detector — the ambient sound was under loud speech or music and never reached the 0.35 bar; 4 are debatable labels (cars visible on screen, traffic noise labelled off-screen); 0 are timing.

- Of the redundant pictures on the development clips, **about 80% are "phantoms"** — sounds the detector reports confidently that are not there at all (a whale at a Christmas market, roaring lions at a helicopter landing). The gate is right that nothing visible makes them; the picture is a detector mistake.

So the **detector** is the bottleneck on both sides, more than the visibility decision.

8. What was tried against the detector, and why it stopped

Each attempt had a pass/fail rule written down before it ran, and was reviewed by an independent AI reviewer ("Fable"). None passed; all are reported honestly.

attempt	idea	result
threshold sweep	try 16 gate settings on the development clips	the current setting was already best
CLAP second opinion	a second audio model (CLAP) must agree with BEATs	CLAP recognises only 50% of real sounds at the loosest useful setting — unusable
PANNs agreement	a second AudioSet detector must agree	removed 24 of 77 phantoms (needed 40); both detectors share the same confusions
persistence + rank rule	count a sound only if it stays among the top-3 for ≥ 1 s, instead of one peak	same trade-off curve as the current rule, just shifted
bigger VLM (32B) for the gate	four times larger vision model	recognises visible sources better (67% vs 35%) but wrongly silences 25% of off-screen sounds (was 3%) — worse under the score
audio language model	ask Qwen2-Audio "which of these sounds do you hear?"	small test passed (6/10 phantoms vetoed, 9/10 real kept); full run removed only 25 of 77
source separation (Demucs)	remove speech before detecting	assessed, not run: removes speech only, and 4 of the 7 masked misses are under music

Conclusion for the thesis: with today's off-the-shelf sound detectors, invented sounds and missed quiet sounds are two ends of one dial; no filter on top fixes

both.

9. Decisions taken and still open

Decided: keep `ONSET_CAM` (a timing refinement measured to make no difference; v3 was evaluated with it on). Skip the independent reference for v3 (it cannot score the gate). Stop detector experiments (§8).

Open for Adam: read the report; decide on submission polish (bibliography entries marked `[verify]`, two figures — pipeline diagram and a per-clip example, three demo videos re-rendered with v3).

Future work (in the report): a detector-only benchmark on the labelled clips; a speech-aware detector measured against it; an audio model with an open vocabulary for sounds AudioSet has no name for ("phone alert"); a second annotator; a study with deaf viewers.

10. Where everything is

- Run one video: `python main.py <video>`; settings in `config.py`.
- Scores: `benchmark/protocol_results_<tag>.json`; `python scripts/paired_stats.py v3 v3_grounded` prints the tables; `scripts/cost_sensitivity.py` the judge-free view.
- Experiment records: `benchmark/gate_setting.json`, `clap_setting.json`, `panns_agree.json`, `persist_rank_setting.json`, `eval_dcase_visibility_32b.json`, `audio_llm_eval.json`.
- Cluster (BIU Slurm, VPN via F5 at `access.biu.ac.il`): `bash slurm/sync_data.sh --code-only` uploads code; jobs are `slurm/*.sh`; the main experiment is `sbatch slurm/job_v3.sh` (~10 h; resumable — resubmit, never restart). A Hugging Face token in `~/.bashrc` on the cluster is needed for the gated models.
- Documents: `docs/report/` (thesis), `docs/project_notes.tex` (lab notebook), `docs/failure_catalogue.md`, `docs/plan_robustness.md` (process rules), `LIMITATIONS.md`.